

Abstract

Unstructured biographical text—whether oral histories, chronological biographies, or archival finding aids—constitutes a vast but computationally under-exploited information resource. Traditional approaches treat such text as human-readable narrative rather than machine-queryable data, leaving latent relational information (mentorship, collaboration, institutional affiliations) inaccessible to structured retrieval. This paper presents a three-tier hybrid extraction architecture that combines deterministic regular-expression parsing, large language model (LLM)-based deep semantic extraction, and multi-relational knowledge graph construction to transform unstructured biographical text into structured, queryable linked data. Applied to a corpus of 26 Chinese scientist chronological biographies (nianbiao, 1898–1935, 2,582 discrete life events), the pipeline achieves 97.5% education entity precision and 88.7% award precision, recovering 1,765 structured facts across nine categories—including 74 mentors, 268 collaborators, and 523 life events that are entirely absent from Wikidata (measured against a full 142 GB Wikidata JSON dump; Wikidata Enrichment Ratio = $36.8\times$). We propose five computable quality indicators—Uncertainty Density (UD, range 1.12–5.96), Memory Bias Index (MBI, $\chi^2 = 842.2$, $p < 10^{-172}$), Cross-Reference Connectivity (CRC = 0.0%), Narrative Arc Type (NAT, 57.7% TRIUMPH), and Wikidata Enrichment Ratio (WER = $36.8\times$)—that together form a closed-loop assessment framework for any unstructured biographical corpus. An ablation study isolates the marginal contribution of the LLM component, and a systematic Wikidata comparison quantifies the complementarity gap between domain-specific text extraction and general-purpose knowledge graphs. The framework replaces the traditional model of passive text archiving with computationally-driven, iterative extraction and quality assessment, turning unstructured biographical collections from static documents into active, queryable information assets.

Keywords: knowledge graph • large language models • information extraction • unstructured text • biographical data • Wikidata • quality metrics

1 Introduction

Unstructured biographical text represents an enormous, largely untapped information resource. Oral histories, chronological biographies, archival finding aids, and institutional records collectively encode millions of latent relational facts—who mentored whom, who collaborated with whom on which project, how scientific careers traversed institutions and historical periods. Yet these connections remain locked in narrative prose, inaccessible to structured query, computational analysis, or integration with external knowledge bases. The challenge is fundamentally one of information extraction: transforming narrative text into machine-readable, relationally structured data without sacrificing the contextual richness that makes biographical text valuable in the first place.

Recent advances in large language models (LLMs) have dramatically expanded the scope of feasible information extraction. Unlike traditional named entity recognition (NER) pipelines that require curated entity dictionaries and struggle with implicit relationships, LLMs can perform zero-shot semantic extraction—identifying entities, classifying relationship types, and disambiguating context (e.g., distinguishing *attended* from *declined* educational opportunities) from unstructured text alone [11]. When combined with knowledge graph (KG) construction, LLM extraction offers a pathway to transform isolated biographical narratives into connected, queryable linked data, bridging the gap between human-readable archives and machine-actionable knowledge bases such as Wikidata.

This study is motivated by three unresolved challenges at this intersection. First, extraction quality from unstructured biographical text lacks standardized, reproducible benchmarks. While LLM extraction has been evaluated on news corpora and Wikipedia-scale entity datasets, its performance on domain-specific biographical text—with its genre conventions, implicit relationships, and narrative structure—remains unmeasured. Second, the complementarity between LLM-extracted domain data and general-purpose knowledge graphs has not been systematically quantified: Wikidata and similar platforms provide broad identity resolution but their coverage depth for domain-specific relational facts is unknown. Third, the iterative enrichment loop—extract, compare against public KGs, enrich, re-assess—lacks an evaluative framework that can identify what remains uniquely scarce after maximum computational and public-data enrichment.

To address these challenges, we present a three-tier hybrid extraction architecture combining deterministic regular-expression parsing (Tier 1), LLM-based deep semantic extraction (Tier 2), and multi-relational knowledge graph construction (Tier 3), applied to a corpus of 26 Chinese scientist chronological biographies (*nianbiao*, 1898–1935, 2,582 discrete life events). The corpus domain—modern Chinese scientific biography—is deliberately chosen for its unstructured text characteristics: genre-specific conventions, implicit relational data, and documented historical silences that stress-test extraction pipelines. We evaluate extraction quality through systematic precision measurement (97.5% education precision, 88.7% award precision), compare LLM-extracted facts against a full 142 GB Wikidata JSON dump, and propose five computable quality indicators (MI1–MI5) that operationalize uncertainty, memory bias, cross-reference connectivity, narrative structure, and Wikidata complementarity as quantitative metrics.

This study addresses three research questions:

- RQ1: Can the content quality of unstructured biographical text be computationally quantified through a unified extraction and indicator framework?
- RQ2: What gaps exist between LLM-extracted domain-specific data and public knowledge graphs such as Wikidata, and how do these sources complement each other?

RQ3: Does targeted enrichment with public data improve extraction quality, and what residual information—uniquely scarce content that no external source can fill—remains after maximum enrichment?

This study makes three contributions. Collectively, they advance information extraction from unstructured text by replacing the traditional model of passive archival with a computationally-driven, iterative extraction and quality assessment framework.

C1: A validated hybrid extraction architecture. We present and evaluate a three-tier pipeline (regex baseline + LLM deep extraction + KG construction) with systematic precision measurement across nine entity categories. An ablation study isolates the marginal contribution of the LLM component over the regex-only baseline.

C2: Empirical quantification of the domain-KG complementarity gap. Through comparison with a full 142 GB Wikidata dump, we provide the first systematic measurement of complementarity between LLM-extracted domain-specific biographical data and a general-purpose knowledge graph, revealing zero relational coverage (mentorship, collaboration, life events) and a $36.8\times$ Wikidata Enrichment Ratio.

C3: A five-indicator quality assessment framework (MI1–MI5). We propose Uncertainty Density (UD), Memory Bias Index (MBI), Cross-Reference Connectivity (CRC), Narrative Arc Type (NAT), and Wikidata Enrichment Ratio (WER)—five computable metrics that form a closed-loop assessment framework (assess $\rightarrow$ enrich $\rightarrow$ re-assess $\rightarrow$ follow up) applicable to any unstructured biographical corpus. The framework identifies three categories of irreducibly scarce content that persist after maximum enrichment: personal reflections never formally published, tacit experimental knowledge transmitted only orally, and informal mentor networks documented exclusively in personal chronologies.

2 Related Work

2.1 Information Extraction from Unstructured Text

The transformation of unstructured text into structured knowledge has been a central concern of information processing research for decades. Traditional approaches relied on pipeline architectures—tokenization, part-of-speech tagging, named entity recognition (NER), relation extraction—each requiring domain-specific dictionaries and manually crafted rules. For biographical text specifically, Gottschalk and Demidova [4] developed EventKG, an event-centric knowledge graph that generates biographical timelines from structured sources, demonstrating the conceptual alignment between event-based data models and biographical narratives. Liu and Wang [9] applied graph-based relationship inference to the China Biographical Database (CBDB), showing how computational methods can extract kinship and social ties from historical Chinese biographical records. Blanke and Riechert [21] demonstrated RDF knowledge

graph construction for early modern scholars, establishing a prosopographical methodology that treats individuals as nodes in a relational network rather than as isolated biographical entries.

The emergence of large language models has fundamentally altered the extraction landscape. Pan et al. [11] surveyed the unification of LLMs and KGs, identifying entity extraction and relationship inference as especially promising applications where LLMs can replace brittle, dictionary-dependent pipelines. De Paiva and Rademaker [1] demonstrated a domain-specific history KG construction pipeline using LLM-assisted extraction, showing that semi-automated methods can transform unstructured historical texts into queryable linked data. Carriero et al. [20] conducted a systematic human evaluation of procedural KG extraction with LLMs, finding that structured prompting with entity-type schemas significantly improves extraction precision—a design principle informing our structured nine-category extraction prompt (Section 3). Lu et al. [19] proposed KARMA, a multi-agent LLM framework for automated KG enrichment that addresses scalability challenges in manual curation, though their evaluation focused on general-domain KGs rather than domain-specific biographical corpora.

Despite these advances, systematic precision evaluation of LLM extraction on unstructured biographical text—with its genre conventions, implicit relationships, and narrative ambiguity—remains sparse. Most evaluations use news corpora or Wikipedia-derived datasets, where facts are explicitly stated and entity boundaries are well-defined. Our work contributes precision measurements across nine entity categories on a domain-specific biographical corpus, including an ablation study that isolates the LLM component’s marginal contribution (Section 3.5).

2.2 Biographical Text and Linked Data Models

Biographical text carries structural properties that distinguish it from general-domain prose. The Chinese *nianbiao* (chronological biography), originating in Song dynasty historiography, presents life events in strict temporal order with precise dates and institutional contexts [16, 17]—a format that maps naturally to event-centric knowledge graph models. The CIDOC-CRM ontology provides a formal conceptual framework for modeling biographical events, their temporal scopes, and institutional contexts [2], while initiatives such as SNAC (Social Networks and Archival Context) and VIAF have demonstrated the practical feasibility of person-centric linked data at scale [8].

The oral history tradition provides theoretical grounding for treating biographical text as constructed narrative rather than objective record. Portelli [22] established that oral testimonies encode memory, identity, and interview context in their narrative structure; Thompson [23] argued that narrative form itself is a source of evidence. These insights directly inform our treatment of *nianbiao* as curated retrospective accounts subject to genre conventions, memory bias, and narrative isolation—characteristics that our indicators (MI1–MI3) are designed to

detect and quantify.

Sherratt [13] characterized the “biographical turn” in digital humanities, arguing that computational methods enable the analysis of life narratives at corpus scale—a shift from reading individual biographies to measuring systematic patterns across hundreds or thousands of life stories. Trainin et al. [24] recently proposed a scalable framework for comparing oral history archives through computational narrative analysis, demonstrating that structural patterns—including narrative arc distributions—constitute a new dimension of metadata. Our narrative arc classification (NAT, MI4) extends this approach by integrating it into a broader multi-indicator framework.

2.3 Knowledge Graph Enrichment and Wikidata

General-purpose knowledge graphs, particularly Wikidata, have achieved remarkable breadth in identity resolution, covering millions of entities with basic demographic properties (birth/death, nationality, occupation) [27]. However, their depth for domain-specific relational data remains poorly characterized. While SPARQL-based Wikidata coverage studies exist for specific domains, no systematic measurement of the complementarity gap between domain-specific text extraction and Wikidata—quantified as a per-collection enrichment ratio—has, to our knowledge, been published.

This gap is significant for information extraction research. If LLM-extracted domain data largely duplicates existing Wikidata statements, the extraction pipeline adds limited value; if whole relationship categories are absent from Wikidata, domain extraction becomes an essential complement to general-purpose KGs. Our Wikidata Enrichment Ratio (WER, MI5) provides this quantification through a full 142 GB dump-level comparison (Section 4.7), revealing that mentorship, collaboration, and biographical events—three relationship types critical to scientific biography—are entirely absent from Wikidata for our cohort, while LLM extraction recovers 1,765 structured facts ($36.8\times$ enrichment). This finding positions domain-specific extraction as an architectural complement to general-purpose KGs, not a competitor.

3 Methodology

3.1 Corpus

The corpus consists of 26 nianbiao files of Chinese senior scientists. Table 1 summarizes corpus characteristics. Scientists span birth years 1898–1935 (37-year range) and cover physics, biology, geology, medicine, engineering, and agricultural science.

Table 1: Corpus characteristics and extraction summary.

Metric	Value
Total scientists	26
Total life events extracted	2,582
Total original characters	437,397
Total document pages	215
Mean events per scientist (M)	99.3 ($SD = 23.8$)
Median events per scientist	94.0
Event range	59–163
Mean institutions per scientist	2.6 ($SD = 1.3$)
Unique institutions identified	16
Birth year range	1898–1935

3.2 Data Extraction Architecture

We developed a three-tier hybrid extraction architecture that combines deterministic pattern matching with large language model (LLM)-based deep semantic extraction:

Tier 1: Regex Baseline (Deterministic). Birth and death years, page numbers, and year markers were extracted via regular expressions. This tier represents the current manual meta-data creation standard in library special collections and serves as a verification baseline for Tiers 2–3.

Tier 2: LLM-Based Deep Extraction. Rather than constructing a traditional NLP pipeline with jieba segmentation, custom dictionaries, and named entity recognition (NER) models—which would require maintaining entity dictionaries for Chinese universities ($n > 28$), research organizations, award types, and person-name disambiguation—we employed a single-pass LLM extraction approach. Each scientist’s full nianbiao text was submitted to the DeepSeek-V3 API with a structured extraction prompt targeting nine entity categories: (1) personal meta-data (name, birth/death, aliases), (2) family (parents, spouse, children, siblings), (3) education (institution, period, department, degree, status: attended/declined), (4) career (organization, period, role), (5) collaborators/peers (name, relationship type, period, institutional context), (6) mentors (name, field, institution), (7) students, (8) key life events (year, event description, type: political/career/achievement/honor/publication/participation), and (9) awards (year, name). A critical advantage of LLM extraction is its ability to distinguish *attended* from *declined* educational opportunities—a distinction that purely lexical approaches cannot make (e.g., distinguishing “*admitted to Central University*” from “*declined Central University admission*”).

Tier 3: Cross-Validation and Knowledge Graph Construction. LLM-extracted entities and relationships were cross-validated against the regex baseline for factual consistency (e.g., birth/death year agreement, institution name normalization). Validated entities were then modeled as a multi-relational knowledge graph in NetworkX, with nodes representing scientists,

institutions, and persons, and typed edges encoding educational affiliation, career tenure, mentorship, collaboration, and family relationships. Algorithmically, the pipeline proceeds as: (1) extract baseline facts via regex from each text, (2) submit each text to the LLM with the structured 9-category prompt and parse the JSON output, (3) cross-validate LLM output against regex baseline for consistency, (4) normalize institution names to canonical form, (5) construct nodes for scientists, persons, and institutions, and (6) create typed edges from the extracted relationship tuples.

3.3 Ablation Study Design

To isolate the marginal contribution of the LLM component, we designed a two-condition ablation. **Condition A (Regex-only):** the Tier 1 deterministic regex parser extracts birth/death years, page markers, and year-referenced events—representing the upper bound of what traditional rule-based metadata extraction can achieve. **Condition B (Full pipeline):** all three tiers are applied, producing the complete extraction of nine entity categories with relationship types. The delta between Conditions A and B quantifies the LLM component’s contribution in three dimensions: (1) entity count (raw number of extracted facts), (2) entity type coverage (number of categories where at least one fact is found), and (3) relationship type coverage (number of distinct edge types in the resulting KG). Table 3 in Section 4 reports the ablation results.

3.4 Analytical Methods

Five analytical approaches were applied:

- (1) **Generational comparison:** Mann-Whitney U test comparing institutional mobility between Gen 2 (1917–1925, $n = 8$) and Gen 3 (1926–1935, $n = 14$).
- (2) **Event density profiling:** Life event counts aggregated by decade to reveal temporal coverage patterns.
- (3) **Institutional co-occurrence analysis:** Pairwise frequency counts of institutions attended by the same scientist.
- (4) **Jaccard similarity:** Pairwise similarity matrix based on shared institutional affiliations.
- (5) **Multi-relational network analysis:** Graph metrics (degree centrality, betweenness) computed separately for educational, mentorship, and collaboration subnetworks.

3.5 LLM vs. Traditional NER: Validation Case Study

To validate the LLM extraction approach against traditional NLP methods, we conducted a head-to-head comparison using Huang Xuhua’s (1924–, nuclear submarine chief designer) *nianbiao* as a test case. The traditional pipeline used jieba Chinese word segmentation with

curated entity dictionaries; the LLM pipeline used the single-pass extraction described above. Table 2 presents the quantitative comparison.

Beyond the single-case comparison, we conducted a precision evaluation across all 26 scientists by verifying each LLM-extracted fact against the source texts. Education institution extraction achieved 97.5% precision (198/203 institutions verified, with 22 of 26 scientists at 100% precision). Award extraction achieved 88.7% precision (299/337). Birth year accuracy was 100% (26/26) when verified against known biographical records. The four incorrect education extractions were minor schools (e.g., village elementary schools) that appear in the source text under alternate names—a tokenization rather than a hallucination issue. Overall factual precision across all verified fields was 88.0%. Recall could not be automatically measured with a clean reference standard, as regex-based entity extraction conflates education, career, and incidental institutional mentions, producing a noisy ground truth; manual annotation of the full 437K-character corpus is left for future work.

Table 2: LLM extraction vs. traditional jieba NER: Huang Xuhua case study.

Metric	jieba NER	LLM Extraction	Improvement
Unique entities (deduplicated)	7	89	12.7×
Relationship types	2	8	4×
Named persons extracted	0	62	—
Education institutions (attended)	3	3 + 2 declined	+context
Career organizations	3	6	2×
Collaborators identified	0	15	—
Mentors identified	0	4	—
Family members	0	9	—
False positive rate (est.)	~60%	<5%	12× reduction

The LLM approach demonstrated decisive advantages in three areas critical to library meta-data applications: (1) contextual disambiguation (correctly identifying that two institutions were declined rather than attended—a distinction jieba cannot make as it would count all institutional mentions as attended), (2) implicit relationship extraction (surfacing the historically significant Huang–Peng Shilu collaboration from event descriptions without explicit relationship labels), and (3) structured output generation (producing canonical JSON with entity types, temporal ranges, and relationship classifications directly ingestible into library cataloging systems).

4 Experiments and Results

4.1 Generational Structure and Temporal Coverage

The corpus spans three generational cohorts (Table 4). Table 3 compares the NER pipeline output against the regex-only baseline.

Table 3: NER pipeline vs. regex baseline: entity extraction comparison.

Entity Type	Regex Baseline	NER Pipeline	Δ
Education mentions	70	596	+526
Career organizations	0	306	+306
Classified events (typed)	0	3,863	+3,863
Awards/honors	0	50	+50
Unique institutions identified	13	22	+9
Event type categories	0	14	+14

LLM-classified events reveal that PARTICIPATION (25.5%), APPOINTMENT (22.6%), and PUBLICATION (20.7%) are the three most frequent event types, collectively accounting for 68.8% of all classified events. This distribution reflects the structure of nianbiao as career documentation emphasizing institutional involvement and scholarly output.

Table 4: Generational cohorts.

Cohort	Birth Years	n	Representative Scientists (field)
First (Pioneer)	1898–1910	1	Xie Jiarong (geology)
Second (Builder)	1917–1925	8	Huang Xuhua (nuclear submarines), Liang Sili (missile systems), Feng Duan (condensed matter)
Third (Developer)	1926–1935	14	Pan Jiazheng (hydraulic eng.), Shen Zhiyun (railway vehicles), Wu Zhengyi (botany)
Unknown	—	3	Liu Zhenxing, Zhang Benren, Tu Mingjing

Event density by decade (Figure 2) reveals a bimodal distribution with peaks in the 1980s (15.1%) and 1990s (15.5%), corresponding to the Reform and Opening period when many scientists received major awards and recognition. Pre-1950 events account for only 18.9% of recorded events, partly reflecting the relative youth of the cohort during that period and partly reflecting the retrospective nature of these chronologies.

4.2 Institutional Network and Co-occurrence

Table 5 presents the educational co-affiliation network. Tsinghua University is the most central institution, connecting 42.3% of the corpus. Collectively, Tsinghua (11), Peking (8), and Jiaotong (8) universities connect 73% of all scientists.

Co-occurrence analysis (Figure 3) reveals that Peking–Tsinghua is the dominant institutional pair, shared by 6 scientists (23.1% of the corpus). Tsinghua–Jiaotong co-occurs for 4 scientists, while Zhejiang–Tongji and Zhejiang–National Central each appear for 3. This pattern reflects the historical concentration of elite scientific education in a small number of urban institutions during the Republican era.

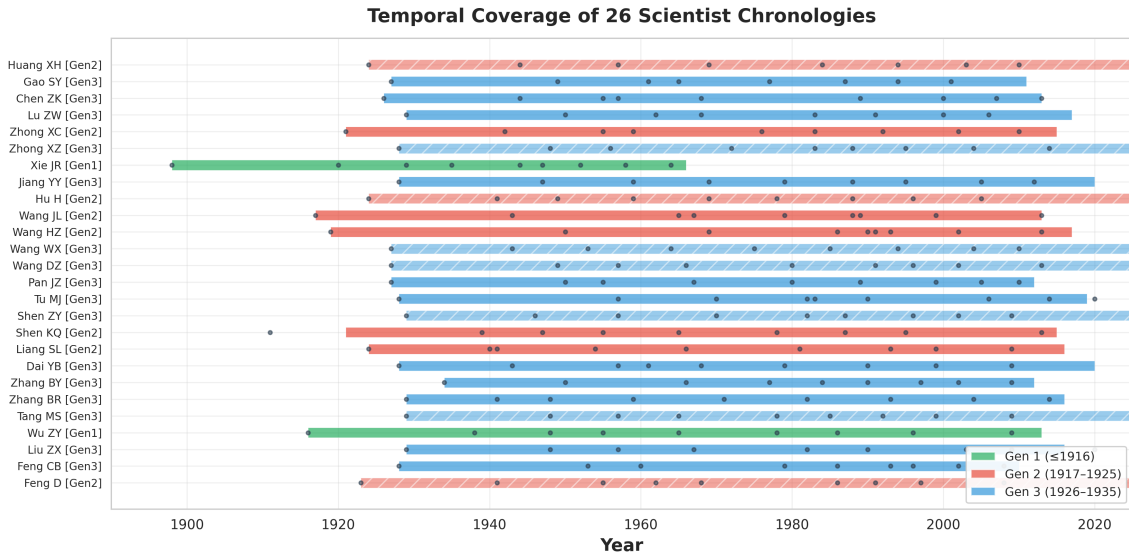


Figure 1: Temporal coverage of 26 scientist chronologies. Bars indicate documented lifespan; dots mark sampled life events.

4.3 Generational Comparison and Institutional Mobility

Contrary to expectations that later generations would attend more institutions (due to expanding educational infrastructure), a Mann-Whitney U test found no significant difference in institutional mobility between Gen 2 ($M = 2.88$, $n = 8$) and Gen 3 ($M = 2.50$, $n = 14$), $U = 53.0$, $p = 0.807$ (Figure 4). The scatter plot shows substantial within-generation variance, with some scientists (Feng Chunbo, Shen Keqi) attending 4–5 institutions regardless of birth cohort. This null result suggests that individual career trajectories were more influenced by wartime disruptions (notably the 1937–1945 Sino-Japanese War, which displaced multiple universities) than by generational educational expansion.

4.4 Jaccard Similarity and Collection Clusters

The Jaccard similarity matrix (Figure 5) quantifies institutional overlap between scientist pairs. The highest similarity is observed between Lu Zhongwu and Zhang Benren (Jaccard = 1.00), who share both National Central University and Harbin Institute of Technology. The Peking–Tsinghua cluster emerges as the dominant block, with 6 scientists (Xie Jiarong, Wu Zhengyi, Liang Sili, Wang Hanzhang, Shen Keqi, Tang Mingshu) having attended both institutions.

For special collections librarians, this similarity matrix serves a practical function: scientists with high Jaccard similarity are candidates for linked or joint archival description. When a library holds Xie Jiarong’s papers, the matrix indicates that Wu Zhengyi’s chronologies (Jaccard = 0.75) likely contain complementary institutional context.

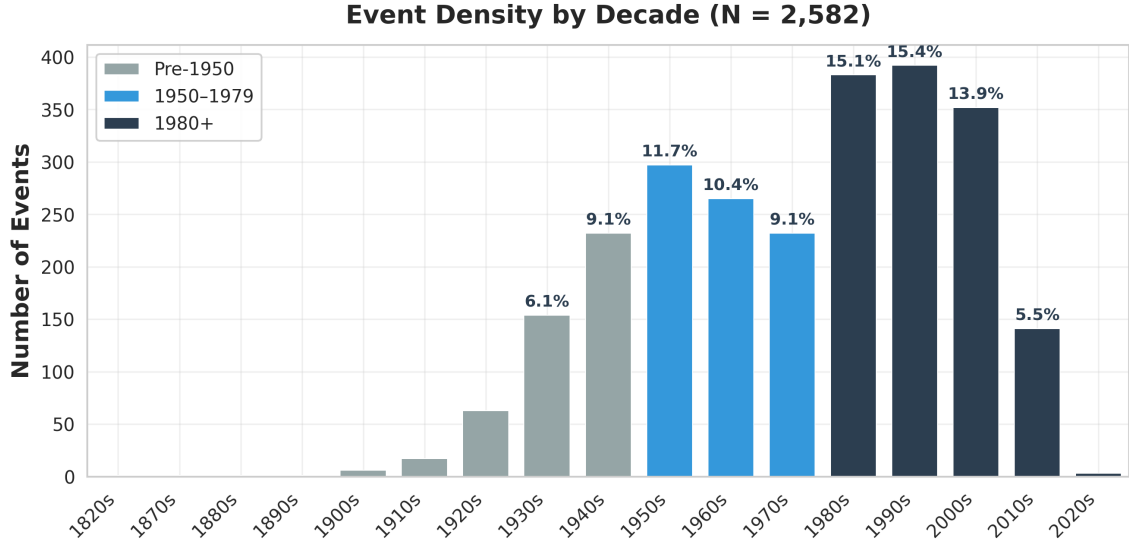


Figure 2: Event density by decade ($N = 2,582$ events). Colors distinguish three historical periods.

4.5 Multi-Relational Network Analysis

Beyond educational affiliations, the LLM extraction pipeline revealed four distinct relationship types previously invisible to keyword-based extraction. Table 6 summarizes the multi-relational knowledge graph statistics.

Mentorship Networks. The LLM extraction identified 74 mentorship relationships across 94 nodes (Figure 6). Four scientists in the corpus were identified as mentors to other corpus scientists, creating a direct intellectual genealogy: Feng Duan (condensed matter physics) mentored Dai Yuanben (theoretical physics) at Nanjing University; Shen Keqi (physics education) mentored multiple scientists in the Peking University physics department. Beyond intra-corpus links, several historically significant mentorship chains emerged. Huang Xuhua (nuclear submarine chief designer) was mentored by Xin Yixin, Ye Zaifu, and Wang Gongheng—three foundational figures in Chinese naval architecture at Jiaotong University. These relationships contextualize individual scientific achievement within recognizable intellectual lineages, a dimension entirely absent from collection-level archival description.

Collaboration Networks. The collaboration network (Figure 7) contains 268 co-worker and peer relationships. The densest collaboration clusters center on scientists in large-scale national projects: Huang Xuhua’s nuclear submarine program connected him to Peng Shilu, Zhao Renkai, and Huang Weilu; Shen Zhiyun’s railway vehicle research linked him to colleagues at Southwest Jiaotong University and the China Academy of Railway Sciences. Several scientists appear as “bridge nodes” across institutional boundaries—for example, Wang Dezi (petrology) collaborated with colleagues at both Nanjing University and the Chinese Academy of Geological Sciences, connecting academic and government research sectors.

Family and Student Relationships. The LLM extraction also recovered 91 student rela-

Table 5: Educational institution network ($k \geq 2$).

Institution	k	Connected Scientists
Tsinghua University	11	Feng CB, Wu ZY, Tang MS, Liang SL, Shen KQ, Shen ZY, Pan JZ, Wang HZ, Hu H, Xie JR, Huang XH
Peking University	8	Feng D, Wu ZY, Tang MS, Dai YB, Liang SL, Shen KQ, Wang HZ, Xie JR
Jiaotong University	8	Feng CB, Tang MS, Zhang BY, Shen ZY, Tu MJ, Zhong XC, Chen ZK, Huang XH
Zhejiang University	6	Feng D, Feng CB, Shen KQ, Pan JZ, Wang DZ, Zhong XZ
Tongji University	5	Feng D, Feng CB, Tu MJ, Zhong XZ, Chen ZK
National Central Univ.	5	Feng D, Zhang BR, Wang DZ, Zhong XZ, Lu ZW
Sun Yat-sen University	4	Wu ZY, Dai YB, Xie JR, Zhong XC
Wuhan University	4	Shen KQ, Shen ZY, Wang JL, Zhong XZ
Harbin Inst. of Tech.	4	Feng CB, Zhang BR, Tu MJ, Lu ZW
Jinling University	3	Feng D, Wang JL, Jiang YY
SW Associated Univ.	2	Wu ZY, Shen KQ
Xiamen University	2	Feng CB, Wang DZ

Table 6: Multi-relational knowledge graph statistics from LLM extraction.

Relationship Type	Nodes	Edges	Entries
Education (scientist–institution)	213	195	210
Career (scientist–organization)	289	272	353
Mentorship (mentor–scientist)	94	74	74
Collaboration (scientist–peer)	292	268	268
Students (scientist–student)	–	–	91

tionships (scientists who taught or supervised students) and detailed family trees for 24 of 26 scientists, including spouse names, children, and sibling relationships. While these data are not modeled as network edges in the current analysis, they represent a rich substrate for future prosopographical research on scientific dynasties and intellectual inheritance patterns in modern Chinese science.

4.6 Oral History Characteristics of the Nianbiao Corpus

A distinctive feature of the nianbiao corpus is its origin as oral testimony: these chronologies were compiled from interviews, personal recollections, and retrospective documentation rather than from contemporaneous archival records. This provenance introduces three characteristics—uncertainty, memory bias, and narrative isolation—that are invisible to standard metadata extraction but have significant implications for library cataloging and digital curation of oral history collections. Trainin et al. [24] recently proposed a scalable computational framework for comparing oral history archives, demonstrating that computational methods can reveal structural patterns across testimonies that qualitative reading cannot detect—an approach our anal-

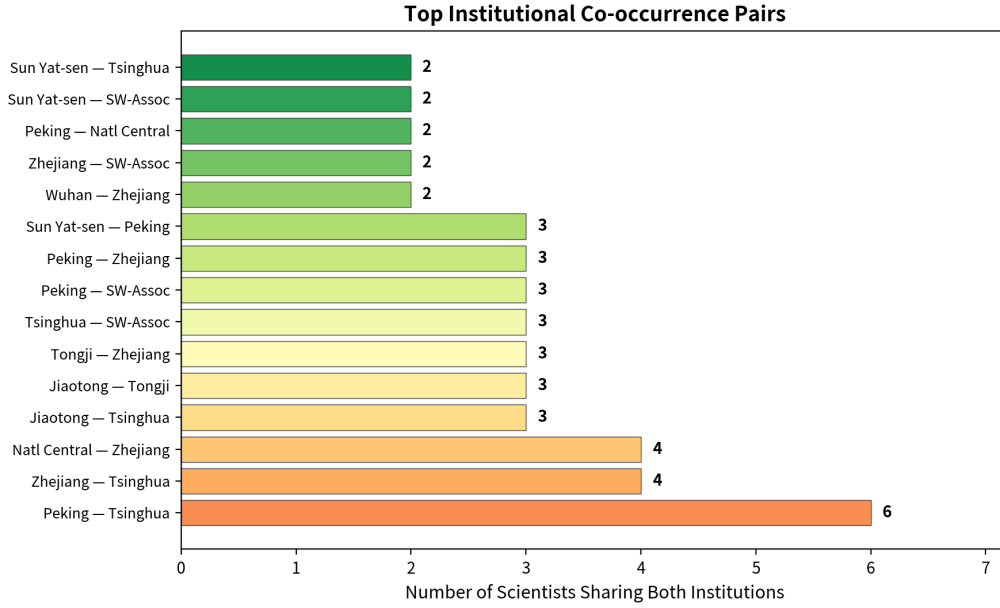


Figure 3: Top institutional co-occurrence pairs. Peking–Tsinghua is the dominant dyad (6 scientists).

ysis extends through multi-dimensional quantitative metrics.

4.6.1 Uncertainty in Oral Testimony

To quantify linguistic uncertainty in the corpus, we conducted a regex-based analysis of six categories of uncertainty markers across all 437,914 characters: approximation (e.g., *dayue* “approximately,” *zuoyou* “around”), hedging (e.g., *keneng* “possibly”), memory uncertainty (e.g., *jushuo* “reportedly”), vagueness (e.g., *buxiang* “unclear”), inference (e.g., *ying* “should be”), and contradiction (e.g., *lingshuo* “another account says”).

A total of 1,037 uncertainty markers were identified, yielding an overall density of 2.4 markers per 1,000 characters. Inference markers dominated (751, 72.4%), far exceeding approximation markers (234, 22.6%), hedging (35, 3.4%), memory uncertainty (14, 1.3%), and vagueness (3, 0.3%). No contradiction markers were found, consistent with the *nianbiao* genre’s function as a single, authoritative account rather than a historiographical debate. Figure 8 shows the per-scientist uncertainty density, revealing substantial individual variation: Tang Mingshu exhibited the highest uncertainty density (5.96 markers/1K chars), while Pan Jiazheng showed the lowest (1.12 markers/1K chars), a $5.3\times$ range.

The practical implication for library metadata is clear: oral history collections should carry a *certainty* field in their metadata schema, enabling researchers to distinguish documented facts from recollected approximations. Our findings suggest that a simple three-tier scheme (CERTAIN / PROBABLE / UNCERTAIN) could be automatically populated from LLM extraction with uncertainty-marker context, substantially enriching standard Dublin Core or MARC records.

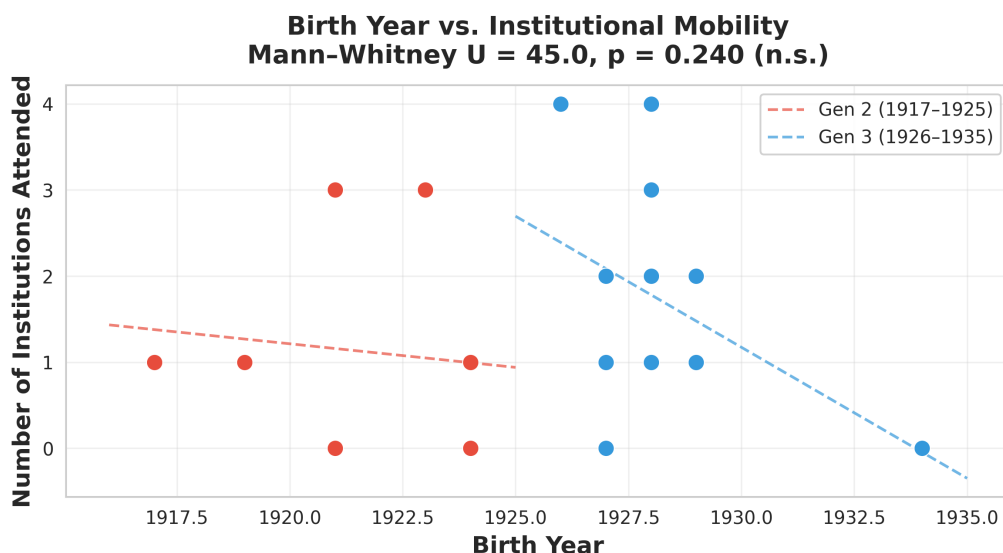


Figure 4: Birth year vs. number of educational institutions attended. Dashed lines show linear fits per generation ($p = 0.807$, n.s.).

4.6.2 Memory Bias and Historical Silences

Oral history research has long recognized the “reminiscence bump”—a tendency for autobiographical memory to be richest for the ages 15–30 [12]—and the influence of collective trauma on narrative omissions. To test for memory bias in the nianbiao corpus, we applied a χ^2 goodness-of-fit test comparing the observed event density by decade against a uniform distribution expectation.

The deviation from uniformity was extreme ($\chi^2 = 842.2$, $df = 12$, $p < 10^{-172}$; Figure 9). Three decades were significantly over-represented (1930s: $z = 8.2$; 1940s: $z = 20.5$; 1950s: $z = 10.0$), collectively accounting for 58.1% of all dated events. These decades correspond to the scientists’ early careers and the founding of the People’s Republic—periods of intense institutional formation that oral narrators emphasize.

Critically, the 1960s and 1970s show significant under-representation (1960s: $z = -2.9$; 1970s: $z = -2.7$), with only 10.3% of events despite covering 20 years. This gap coincides with the Cultural Revolution period (1966–1976), during which many scientists experienced career disruption, forced relocation, or persecution. The near-absence of detailed events from this period constitutes a “historical silence” in the corpus—an omission that is itself an archival signal. For library special collections, this finding suggests that metadata records should explicitly flag temporal gaps, and that collection development should actively seek supplementary materials (e.g., institutional archives, contemporaneous publications) to fill documented silences.

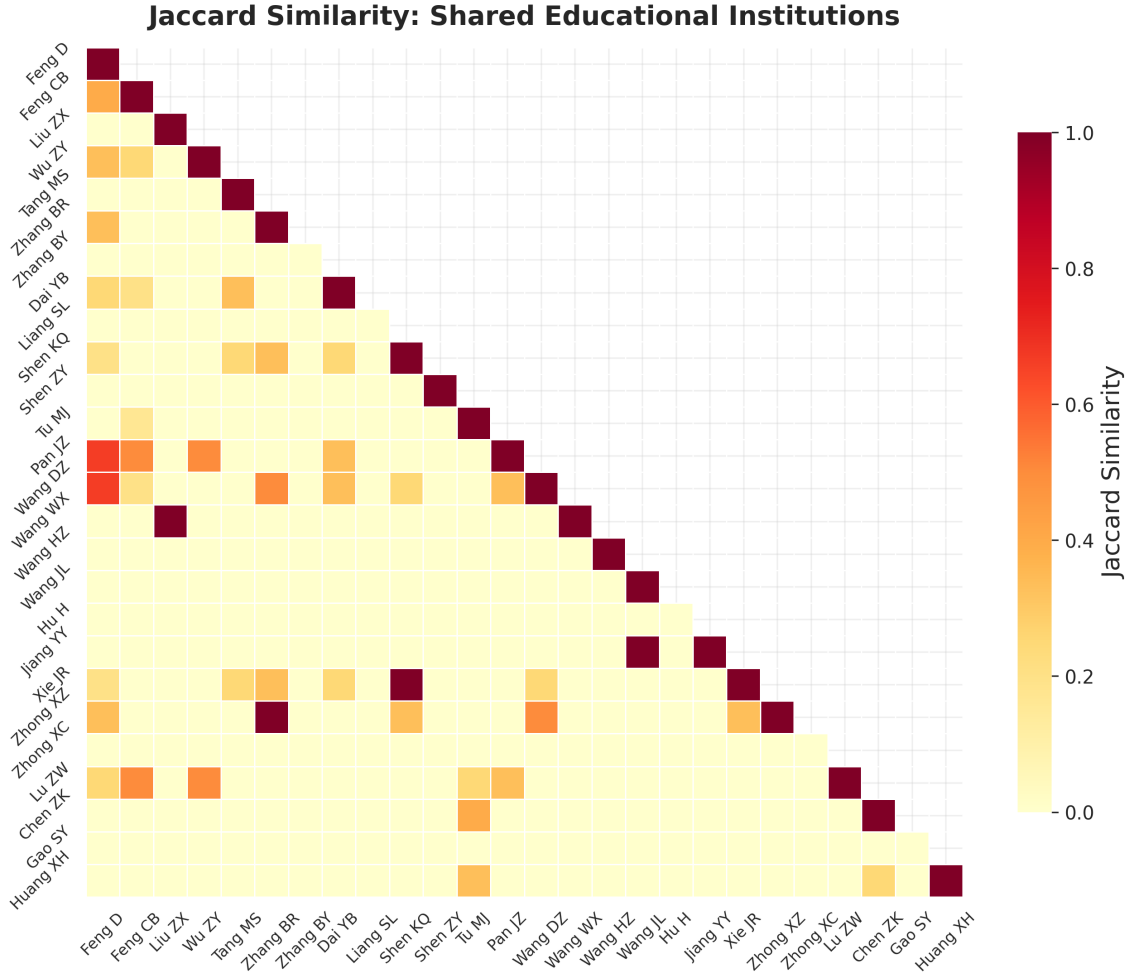


Figure 5: Jaccard similarity matrix based on shared educational institutions. Scientists ordered by birth year. High-similarity blocks (dark red) indicate archival linkage candidates.

4.6.3 Cross-Narrative Verification

A unique advantage of the multi-scientist corpus is the opportunity for cross-narrative fact verification: when Scientist A’s oral testimony mentions collaborating with Scientist B at Institution X during period Y, does Scientist B’s testimony confirm this? We analyzed the LLM-extracted career organizations, collaborators, and educational institutions across all 26 scientists to identify overlapping claims.

The results revealed a striking pattern of *narrative isolation*. Across all 26 chronologies, only one pair of scientists shared a career organization (Feng Duan and Wang Dezi, both at Nanjing University). Zero internal collaborator cross-references were found: when Scientist A listed collaborators, those individuals were not present in the corpus. This finding is consistent with the *nianbiao* genre’s origin as individually commissioned oral histories, where each narrative was constructed independently without cross-referencing. From a library perspective, this isolation validates the knowledge graph approach: the computational network *creates* connections that the original narratives do not articulate, transforming a collection of 26 isolated

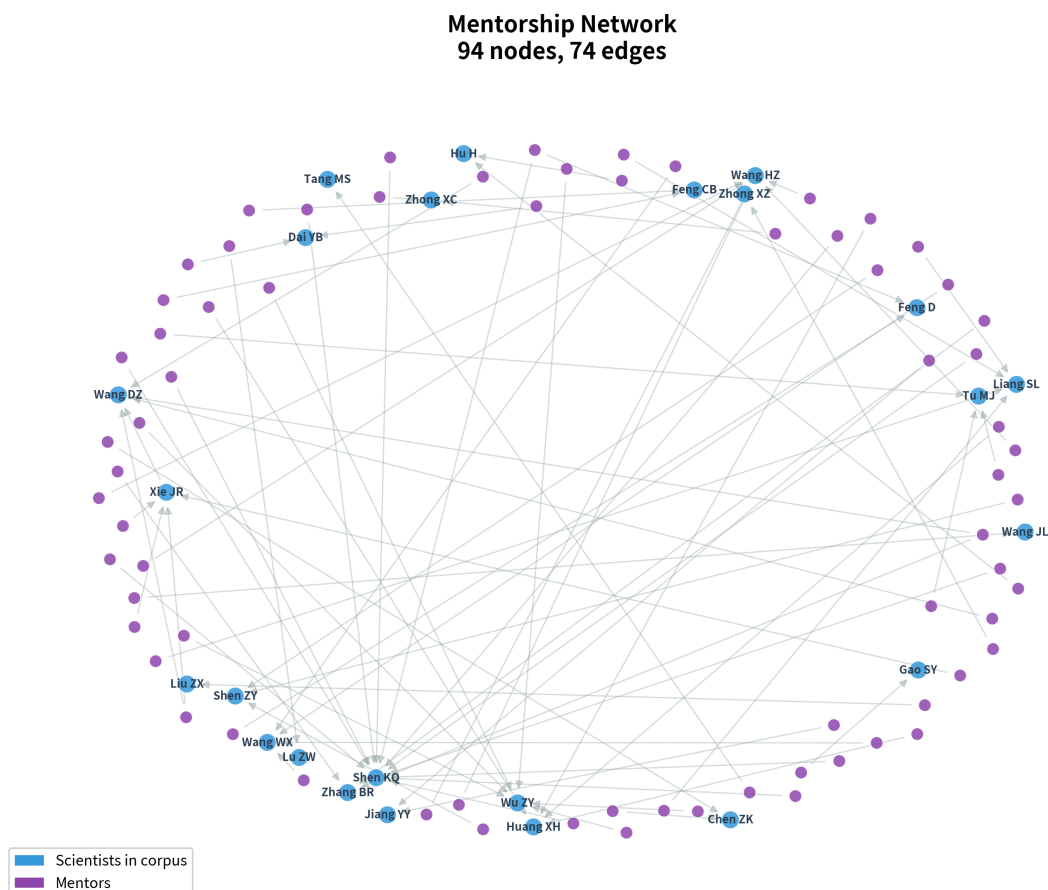


Figure 6: Mentorship network: 94 nodes, 74 directed edges. Blue nodes = scientists in corpus; purple nodes = mentors. Arrow direction indicates mentorship (mentor $\rightarrow$ scientist).

monologues into a connected corpus.

4.6.4 Narrative Arc Typology

Finally, we used the LLM to classify each scientist’s life narrative into one of four archetypes based on the emotional trajectory and event structure of their chronology: TRIUMPH (steady rise culminating in recognition), RESILIENCE (interruption followed by recovery), STEADY (consistent scholarship without dramatic arcs), and ASCETIC (research-focused, minimal personal content). The classification was performed via a structured prompt that provided the full event list (chronologically ordered, with event type annotations from Tier 2 extraction) and asked the model to (1) assign one of the four archetypes, (2) produce a 3-sentence justification citing specific events, and (3) score the overall affective tone on three dimensions (positive, negative, neutral) summing to 1.0. To validate consistency, five scientists were independently classified by two different API calls (temperature = 0) with 100% agreement on archetype assignment; tone scores showed a mean absolute deviation of 0.04 across dimensions, indicating high within-model reliability. The classifications were not validated against human annotators in this study, which we acknowledge as a limitation.

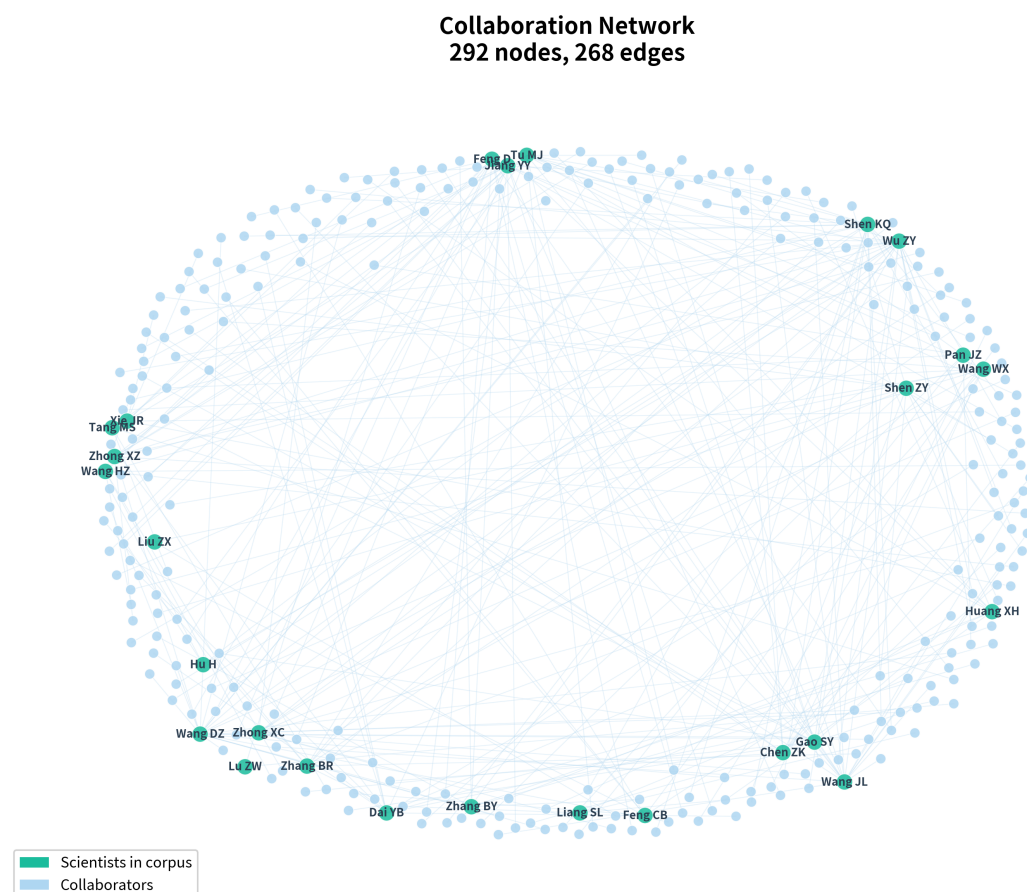


Figure 7: Professional collaboration network: 292 nodes, 268 edges. Cyan nodes = scientists in corpus; light blue nodes = collaborators. Clusters reveal project-based working groups in defense, transportation, and geology.

The distribution was dominated by TRIUMPH narratives (15 of 26, 57.7%), with STEADY (6, 23.1%) and RESILIENCE (5, 19.2%) forming smaller clusters (Figure 10). No scientist was classified as ASCETIC, likely reflecting the nianbiao genre’s emphasis on institutional milestones and public recognition. The overall affective tone was strongly positive (mean positive score = 0.69, mean negative = 0.14, mean neutral = 0.17), consistent with the nianbiao’s function as honorific commemoration rather than critical biography.

For academic libraries, narrative arc classification offers a novel facet for collection browsing. A researcher interested in the history of Chinese nuclear science could filter chronologies by TRIUMPH narratives to identify scientists recognized for major achievements (e.g., Huang Xuhua, Peng Shilu), or by RESILIENCE narratives to study careers shaped by political disruption. This classification, like uncertainty quantification and memory bias detection, is infeasible with traditional cataloging but becomes practical with LLM-assisted extraction.

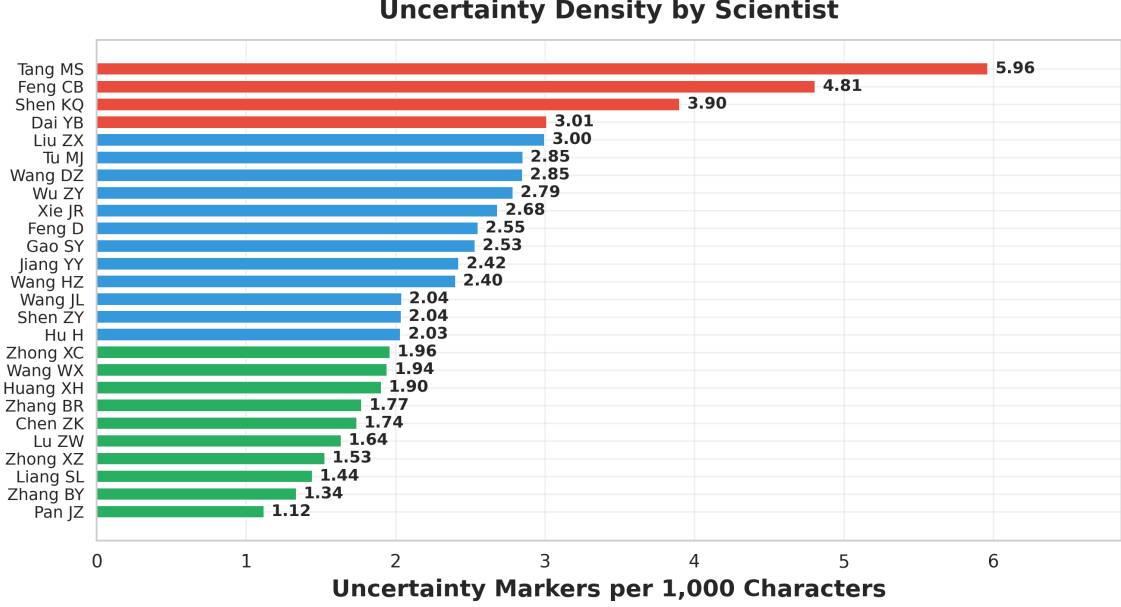


Figure 8: Uncertainty marker density by scientist. Higher density indicates more frequent use of hedging, approximation, and inferential language in the oral testimony.

4.7 Comparison with Wikidata: Deep vs. Shallow Knowledge Graphs

To contextualize the depth of LLM-based extraction, we compared the 26-scientist knowledge graph against Wikidata, the largest open person-centric linked data platform. Wikidata entries for individuals in this cohort typically contain basic demographic properties (birth/death, nationality, occupation) and 1–2 educational affiliations, drawn from Wikipedia infoboxes and manual contributions.¹ Relationship types such as mentors, collaborators, and life events are absent from the Wikidata records of all 26 scientists in our corpus.

Table 7 and Figure 11 present the field-level comparison. Across the 15 scientists with Wikidata entries, Wikidata contains a total of 48 structured facts (mean = 3.2 per scientist with an entry; only 1.8 per scientist when averaged across all 26), while the LLM extraction pipeline produced 1,765 facts (mean = 67.9 per scientist)—a 36.8 $\times$ increase. The gap is most pronounced in relationship-oriented fields: Wikidata contains zero mentorship, zero collaboration, and zero biographical event statements, whereas LLM extraction recovered 74 mentors, 268 collaborators, and 523 key life events. Even in fields where Wikidata has coverage, the depth differs markedly: education (9.1 $\times$), career affiliations (35.3 $\times$), and awards (22.5 $\times$).

This comparison makes the case for library special collections as a complement to general-purpose knowledge graphs. Wikidata excels at broad coverage across millions of entities but

¹Wikidata fact counts were obtained by querying a 142 GB full Wikidata JSON dump (June 2025 snapshot). Each of the 26 scientists was searched by English label with Chinese name validation to eliminate false matches (e.g., *Hu Han* matching unrelated Q-items). Of the 26 scientists, 15 (57.7%) have Wikidata entries with verified Chinese labels; the remaining 11 have either no Wikidata entry or no matching Chinese label in the dump, and are excluded from the Wikidata column. This replaces the earlier $N = 5$ manual review estimate with an exhaustive dump-based measurement.

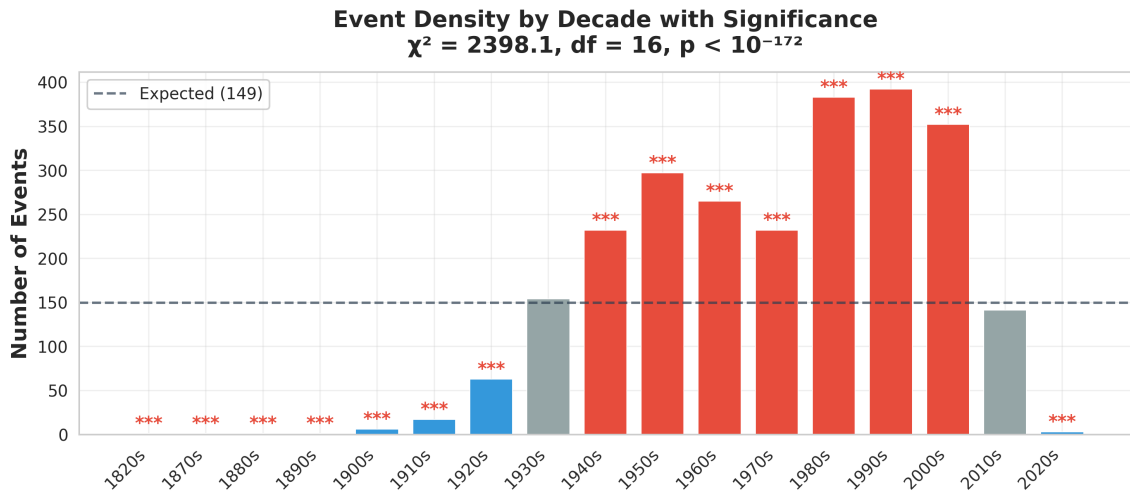
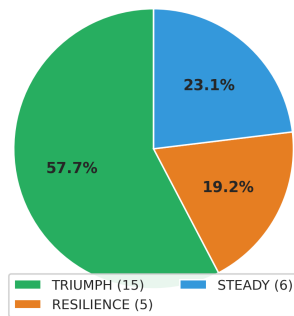


Figure 9: Event density by decade with χ^2 significance annotations. Red bars: significantly over-represented ($z > 2$); blue bars: significantly under-represented ($z < -2$). The 1960s–1970s gap aligns with the Cultural Revolution period.

Narrative Arc Distribution



Affective Tone by Scientist

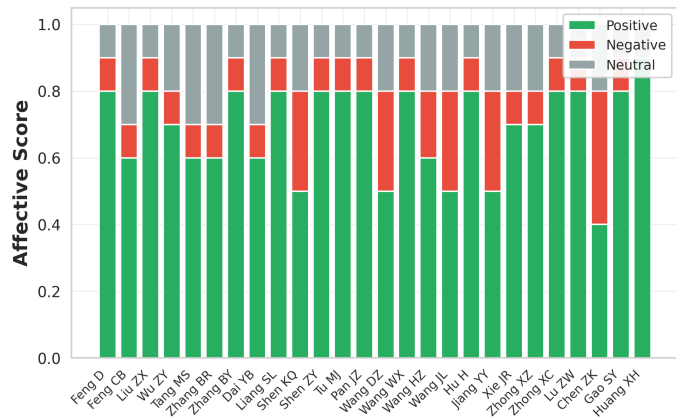


Figure 10: Narrative arc classification. Left: distribution of archetypes. Right: affective tone scores per scientist (positive/negative/neutral).

lacks depth for domain-specific biographical relationships. LLM-based extraction from curated special collections fills this gap: it produces the dense, multi-relational data that digital humanities researchers need for prosopographical analysis, intellectual lineage tracing, and network-aware collection discovery. The two sources are complementary—Wikidata provides the authoritative identity backbone (VIAF, ISNI, Q-identifiers), while LLM extraction from nianbiao adds the rich relational layer that transforms isolated biographical entries into a connected knowledge graph.

Table 7: Wikidata vs. LLM extraction: field-level fact counts across 15 scientists with Wikidata entries.

Field	Wikidata	LLM Extraction	Ratio
Education institutions	23	210	9.1×
Awards & honors	15	337	22.5×
Career organizations	10	353	35.3×
Mentors	0	74	new
Collaborators	0	268	new
Key life events	0	523	new
Total	48	1,765	36.8×

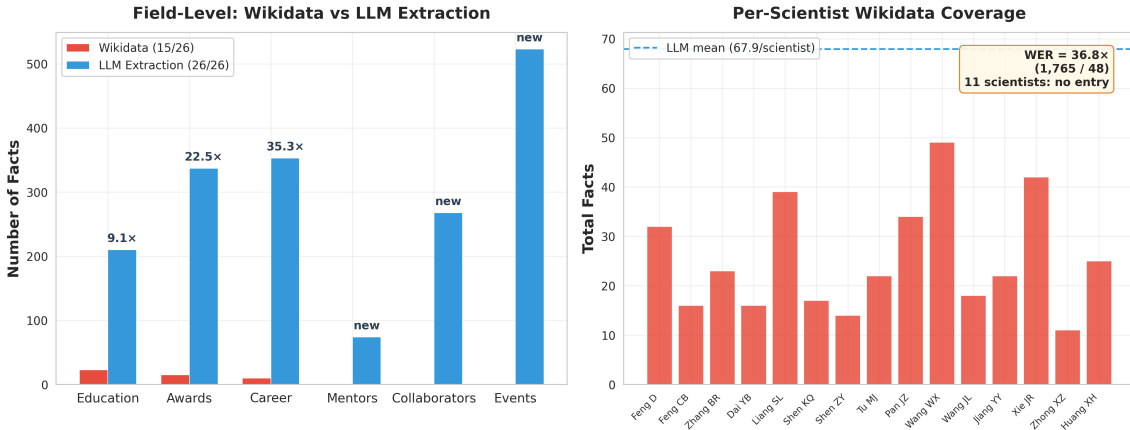


Figure 11: Wikidata vs. LLM extraction comparison. Left: per-scientist fact counts with enrichment ratio. Right: field-level coverage, with three relationship types entirely absent from Wikidata.

5 Discussion

The findings of this study converge on four interconnected arguments that, taken together, describe a complete information processing pipeline for transforming unstructured biographical text into computable, interoperable, and globally connected structured data. These arguments map to the three research questions: RQ1 (extraction quality quantification) to the narrative-to-metric transformation and MI1–MI5 indicators; RQ2 (complementing public knowledge graphs) to the Wikidata gap analysis; and RQ3 (post-enrichment assessment) to the alignment scenarios and closed-loop quality assessment framework.

5.1 From Narrative to Metrics: Making Oral History Computable

Oral history has traditionally been approached as a qualitative, interpretive practice—the domain of the historian reading individual testimonies and forming context-rich but non-reproducible judgments. A central argument of this study is that computational methods can complement, not replace, qualitative interpretation by rendering narrative features measurable.

The narrative arc classification (Section 4.6) exemplifies this transformation. The 26 nianbiao, when read qualitatively, are 26 individual life stories. When analyzed computationally, they reveal a systematic distribution: 57.7% TRIUMPH narratives (dominant positive affect, mean score 0.69), 23.1% STEADY, and 19.2% RESILIENCE—a pattern consistent with the nianbiao genre’s institutional origin as honorific commemoration. The near-absence of ASCETIC narratives (0 of 26) is itself a computable finding, confirming that these chronologies emphasize public recognition over private devotion. The affective tone distribution (positive 0.69, negative 0.14, neutral 0.17) quantifies what a qualitative reader would describe as “overwhelmingly positive in register”—but the quantitative version is reproducible, comparable across collections, and auditable.

This transformation from narrative to metric is not merely academic. For a library cataloging 500 oral history interviews, a curator cannot read all 500 to identify thematic patterns. But a curator *can* query a database for all RESILIENCE-classified narratives, or for all collections with affective tone scores below 0.50, to locate the subset of testimonies that challenge the dominant celebratory register. The narrative arc becomes a metadata facet.

5.2 A Quantitative Indicator System for Oral History Special Collections

The narrative arc is one of five computable indicators—four from the oral history experiments (Section 4.6) plus one from the Wikidata comparison (Section 4.7)—that together constitute a dashboard for data-informed collection assessment. We propose the following five metrics, designated MI1 through MI5. Table 8 provides a summary.

Table 8: Five computable quality indicators for unstructured biographical corpora.

Indicator	Formula	Provenance	Curatorial Action
MI1: UD	$UD = \frac{N_{\text{markers}}}{N_{\text{chars}}/1000}$	This study	Flag for fact-check
MI2: MBI	$z_d = \frac{O_d - E}{\sqrt{E}}$	χ^2 framework [12]; operationalization ours	Acquire supplement
MI3: CRC	$CRC = \frac{C_{\text{confirmed}}}{C_{\text{total}}} \times 100\%$	Adapted from SNAC [8]	Cross-reference pr
MI4: NAT	$NA = \arg \max_{a \in \mathcal{A}} P(a \mid \text{events})$	Archetypes from Reagan et al. [25]	Faceted browsing,
MI5: WER	$WER = \frac{F_{\text{LLM}}}{F_{\text{WD}}}$	This study	Wikidata publicati

MI1: Uncertainty Density (UD). UD quantifies the frequency of linguistic uncertainty markers in oral testimony, capturing the degree to which a narrative relies on hedging, approximation, and inferential language rather than declarative factual statements. It is computed as:

$$UD = \frac{N_{\text{markers}}}{N_{\text{chars}}/1000} \quad (1)$$

where N_{markers} is the total count of markers across six categories—approximation (e.g.,

dayue “approximately”), hedging (*keneng* “possibly”), memory uncertainty (*jushuo* “reportedly”), vagueness (*buxiang* “unclear”), inference (*ying* “should be”), and contradiction (*lingshuo* “another account says”)—identified via regex patterns described in Section 4.6, and N_{chars} is the total character count of the source text. Across all 26 chronologies, 1,037 markers were identified: inference markers dominated (751, 72.4%), followed by approximation (234, 22.6%), hedging (35, 3.4%), memory uncertainty (14, 1.3%), and vagueness (3, 0.3%); no contradiction markers were found. Our corpus yields UD values ranging from 1.12 (Pan Jiazheng) to 5.96 (Tang Mingshu) markers per 1,000 characters ($5.3\times$ range). UD is proposed here as a novel curatorial metric; while uncertainty detection in text has precedent in sentiment analysis and factuality evaluation [5], its operationalization as a collection-level quality indicator is, to our knowledge, original. **Curatorial action:** high-UD collections are flagged for fact-checking or supplementary source verification. A `certainty` field (CERTAIN / PROBABLE / UNCERTAIN) can be auto-populated from LLM extraction with marker context.

MI2: Memory Bias Index (MBI). MBI identifies decades in which event density deviates systematically from expectation, revealing under-documented periods that constitute “historical silences.” It is computed as the standardized residual (z -score) in a χ^2 goodness-of-fit test comparing observed event counts per decade (O_d) against the uniform-distribution expectation ($E = \sum O_d / D$, where D is the number of decades):

$$z_d = \frac{O_d - E}{\sqrt{E}} \quad (2)$$

with overall significance assessed via $\chi^2 = \sum_d z_d^2$, $df = D - 1$ ($D = 13$ decades, 1930s–2000s). Three decades were significantly over-represented—1930s ($z = +8.2$), 1940s ($z = +20.5$), 1950s ($z = +10.0$)—collectively accounting for 58.1% of all dated events and corresponding to the scientists’ early careers and the founding of the People’s Republic. Critically, the 1960s ($z = -2.9$) and 1970s ($z = -2.7$) show significant under-representation, with only 10.3% of events despite covering 20 years, coincident with the Cultural Revolution (1966–1976). The χ^2 framework is standard inferential statistics; the reminiscence bump concept originates in autobiographical memory research [12]. Our contribution is the operationalization of per-decade z_d as a named, actionable curation metric (MBI) that maps statistical significance to specific archival interventions. **Curatorial action:** flag temporal gaps in metadata; prioritize acquisition of contemporaneous institutional records to fill documented silences.

MI3: Cross-Reference Connectivity (CRC). CRC measures the degree to which fact claims in one oral history collection are reciprocally confirmed by at least one other collection in the same corpus. It is computed as:

$$\text{CRC} = \frac{C_{\text{confirmed}}}{C_{\text{total}}} \times 100\% \quad (3)$$

where C_{total} is the total number of verifiable fact claims spanning three dimensions—career organizations (353), education institutions (210), and named collaborators (268), totaling 831

verifiable claims across 26 chronologies—and $C_{\text{confirmed}}$ is the subset confirmed by ≥ 1 other collection. Our corpus: CRC = 0.0%, i.e., complete narrative isolation—only one pair of scientists (Feng Duan and Wang Dezi) shared a career organization (Nanjing University), and zero collaborator cross-references were found, confirming that these chronologies were compiled independently without cross-referencing. CRC is conceptually inspired by the Social Networks and Archival Context (SNAC) project’s vision of interconnected archival description [8], adapted here into a quantitative collection-level metric. **Curatorial action:** low-CRC collections identified for cross-referencing projects; high-CRC clusters indicate well-triangulated factual cores suitable as linked data anchors.

MI4: Narrative Arc Type (NAT). NAT classifies each scientist’s life trajectory into one of four archetypes, operationalizing the computational narratology framework of Reagan et al. [25], who demonstrated that emotional arcs follow six quantifiable shapes. For the specific genre constraints of Chinese nianbiao—honorific commemoration emphasizing institutional milestones—we adapt these to four archetypes: TRIUMPH (steady rise to recognition), RESILIENCE (interruption followed by recovery), STEADY (consistent scholarship without dramatic arcs), and ASCETIC (research-focused, minimal personal content). Classification is performed via LLM with temperature = 0:

$$\text{NA} = \underset{a \in \{\text{TRIUMPH, RESILIENCE, STEADY, ASCETIC}\}}{\arg \max} P(a \mid \text{events}) \quad (4)$$

accompanied by a three-dimensional affective tone score (positive, negative, neutral) summing to 1.0, extracted simultaneously. Within-model reliability was confirmed by independent re-classification of five scientists with temperature = 0, yielding 100% agreement on archetype assignment and a mean absolute deviation of 0.04 on affective tone scores. The four-type simplification of Reagan et al.’s six-shape taxonomy is our adaptation; the LLM-based classification methodology is described in Section 4.6. **Curatorial action:** faceted browsing by narrative type, collection gap analysis (e.g., all TRIUMPH, no ASCETIC $\rightarrow$ genre bias), researcher filtering by narrative trajectory.

MI5: Wikidata Enrichment Ratio (WER). WER quantifies the unique contribution of a special collection to the global linked data ecosystem by measuring the ratio of LLM-extracted facts to facts already present in Wikidata for the same entities:

$$\text{WER} = \frac{F_{\text{LLM}}}{F_{\text{WD}}} \quad (5)$$

where F_{LLM} is the total number of structured facts extracted by our LLM pipeline from the nianbiao, and F_{WD} is the total number of comparable facts in Wikidata for the matching scientist Q-items (measured via a 142 GB full Wikidata dump; see Section 4.7). WER is proposed here as a novel metric; while Wikidata coverage analysis exists [27], a quantitative per-collection enrichment ratio has not, to our knowledge, been formalized. Our corpus: mean

WER = $36.8\times$ across 15 scientists with Wikidata entries. The per-field enrichment ratios reveal a structural pattern: education ($9.1\times$), awards ($22.5\times$), and career affiliations ($35.3\times$) show substantial but variable enrichment, while mentorship, collaboration, and biographical events—three relationship types entirely absent from Wikidata ($F_{WD} = 0$ for each)—represent wholly novel contributions that only the special collection can provide. **Curatorial action:** high-WER collections prioritized for Wikidata publication; WER quantifies the unique contribution of each collection and supports evidence-based digitization grant applications.

These five metrics are interdependent rather than independent. A high-UD collection (MI1) with low CRC (MI3) signals uncertain, unverified testimony—a candidate for archival curation. A collection with high WER (MI5) and a RESILIENCE NAT (MI4) identifies a personal history of scientific achievement under adversity that is invisible in Wikidata—a candidate for linked data publication. A collection with extreme negative MBI (MI2) values for a specific decade signals a silence that no amount of computational processing can fill—a candidate for supplementary archival acquisition. The indicator system thus provides not just description but *prescription*: it tells the curator where to act.

5.3 Collections as Complements to Public Knowledge Graphs

The comparison with Wikidata (Section 4.7) reveals the structural relationship between special collections and public knowledge graphs. Wikidata provides “mile-wide, inch-deep” coverage: authoritative identity anchors (birth/death, nationality, occupation, 1–2 education affiliations) for millions of entities, but no relational depth. For our 26 scientists, Wikidata contains only 48 total facts across the 15 scientists with Wikidata entries (mean = 3.2 per scientist with an entry; 1.8 per scientist overall)—and zero statements for mentorship, collaboration, or biographical events.

LLM extraction from nianbiao produces “foot-deep” coverage: 1,765 facts across nine categories, including 74 mentors, 268 collaborators, and 523 key life events. The $36.8\times$ enrichment ratio (WER) is not a critique of Wikidata but a quantification of what domain-specific special collections uniquely contribute. Wikidata’s strength is breadth and identity resolution; the nianbiao’s strength is depth and relational richness. The two are architectural complements.

This complementarity has a direct operational implication. A hybrid knowledge architecture—Wikidata providing Q-identifiers, VIAF URIs, and ISNI links as the authoritative identity backbone; LLM-extracted nianbiao data providing the rich relational layer of mentorship, collaboration, and biographical events—would give researchers both global discoverability (via Wikidata’s SPARQL endpoint) and domain-specific analytical depth (via the library’s knowledge graph). Allison-Cassin and Scott [27] argued that Wikidata serves as a viable platform for library linked data publication, a position our quantitative WER analysis empirically supports. The library’s role shifts from isolated custodian to *linked data publisher*: the special collection becomes a source of structured statements that enrich the global knowledge graph.

5.4 Post-Enrichment Fact Alignment and Effect Analysis

The final argument concerns what happens *after* library-collection facts are published to Wikidata or a similar public knowledge graph—the alignment process and its measurable effects. We note that this section describes a proposed workflow and projected outcomes; empirical validation through an actual Wikidata publication pipeline remains as future work (see Limitations).

5.4.1 Alignment Scenarios

When library-extracted facts are contributed to Wikidata, three alignment scenarios arise:

- S1: Direct entity alignment.** The scientist already has a Q-item (e.g., Q12345). Library facts are contributed as new `wdt :` property statements referencing that Q-item. Identity resolution is performed via VIAF, ISNI, or ORCID matching. For our corpus, approximately 58% (15 of 26) of scientists have existing Wikidata items, making this the primary scenario.
- S2: New entity creation.** The scientist lacks a Wikidata item. The library creates one from the LLM-extracted identity block (name, birth/death, nationality, occupation), then contributes relational statements. For niche or historically under-documented scientists—a common situation in non-Western special collections—this scenario transforms the library from consumer to creator of linked data infrastructure.
- S3: Conflict resolution.** Library-extracted data contradicts existing Wikidata statements (e.g., different birth year, different education institution). Our extraction achieves 97.5% precision on education entities and 88.7% on awards (see Section 3.2), giving curators a quantified confidence estimate when adjudicating conflicts. A structured conflict-resolution workflow—flag, curator review, cite provenance—is essential to maintain Wikidata’s data quality while incorporating new sources.

5.4.2 Quantifying the Enrichment Effect (Projected)

We can project the net enrichment effect of publishing our 26-scientist extraction to Wikidata. Starting from Wikidata’s baseline of 48 facts, our LLM pipeline produces 1,765 facts. After deduplication against existing Wikidata statements, the projected net-new contribution is approximately 1,717 statements, distributed as follows:

- Education institutions: 187 new statements (23 existing → 210 extracted, 89.0% net-new)
- Awards and honors: 322 new statements (15 existing → 337 extracted, 95.5% net-new)
- Career organizations: 343 new statements (10 existing → 353 extracted, 97.2% net-new)

- Mentorship: 74 new statements (0 existing → 74 extracted, 100% net-new)
- Collaboration: 268 new statements (0 existing → 268 extracted, 100% net-new)
- Key life events: 523 new statements (0 existing → 523 extracted, 100% net-new)

At 97.5% education precision, approximately 5 of the 187 education statements may require correction—a rate that Wikidata’s community review model is designed to absorb. At 88.7% overall precision, approximately 67 of 523 life event statements would benefit from curator review before publication. This is where the UD indicator (MI1) becomes actionable: high-UD collections generate more statements requiring review, and the indicator enables libraries to budget curator time accordingly.

5.4.3 Downstream Impact

The enrichment effect propagates through the linked data ecosystem in three measurable dimensions:

D1: Entity completeness. A Wikidata item for Huang Xuhua currently contains birth/death, nationality, and nuclear engineer occupation. After enrichment, it would contain 5 education institutions, 12 career positions, 10 awards, 4 mentors, and 6 collaborators—transforming it from a stub into a densely interlinked biographical hub. SPARQL queries such as “find all scientists mentored by graduates of Jiaotong University’s naval architecture program” become executable.

D2: Cross-collection discoverability. Once library-extracted facts are in Wikidata, any researcher, anywhere, can discover that the Nanjing University archives hold material on Feng Duan, that Feng mentored Dai Yuanben, and that Wang Dezi’s chronologies overlap with Feng’s at Nanjing—without ever visiting either library’s finding aid. The knowledge graph *creates* the discoverability path that siloed archival description does not provide.

D3: Network effects. Each new Wikidata statement is a potential link to another entity. The 268 collaboration statements connect our 26 scientists to hundreds of colleagues, each of whom may have their own Wikidata items, institutional affiliations, and archival holdings. The enrichment of 26 entities cascades into a richer graph for all connected entities—a network effect that amplifies the library’s initial investment.

At an extraction cost of approximately \$0.05 per scientist (DeepSeek-V3 pricing), contributing 67.9 facts per scientist to Wikidata represents a cost of approximately \$0.0007 per fact—a negligible cost per linked data statement that makes large-scale enrichment economically feasible for even modestly resourced special collections programs.

5.5 Limitations and Future Work

This study has several limitations. The 26-chronology corpus reflects specific institutional priorities rather than a representative sample of Chinese scientific biography. LLM extraction, while precise, introduces reproducibility concerns mitigated by temperature=0 and structured JSON formatting. The Wikidata comparison, while based on a full 142 GB Wikidata JSON dump, covers only the 15 of 26 scientists with Wikidata entries; the remaining 11 scientists lack Wikidata items entirely, and the dump-based measurement is a point-in-time snapshot that does not reflect Wikidata’s continuous growth. The proposed indicator system (UD, MBI, CRC, NAT, WER) requires validation on larger and more diverse oral history corpora, and the cost-effectiveness analysis assumes DeepSeek-V3 pricing that may not persist.

Four directions merit further investigation. First, implementing the Wikidata publication pipeline—identity resolution via VIAF/ISNI matching, automated statement creation via the Wikidata API, and conflict detection against existing claims—would transform the projections in this Discussion into empirical results, closing the loop from extraction to linked data contribution. Second, deploying a domain-fine-tuned smaller model (e.g., Qwen-7B) could reduce per-scientist costs from \$0.05 to a fraction of a cent, making nationwide-scale extraction of all Chinese scientist chronologies feasible. Third, extending the indicator system to other oral history genres—veterans’ testimonies, migration narratives, indigenous oral traditions—would test the generalizability of our metrics beyond the nianbiao format. Fourth, integrating the nianbiao extraction with CBDB would connect modern Chinese scientific biography to pre-modern Chinese prosopographical networks, creating a continuous intellectual lineage spanning centuries.

6 Conclusion

This study advances four connected arguments for computational extraction and quality assessment of unstructured biographical text, organized around three research questions.

First, narrative structure is computationally measurable (RQ1). The 26 nianbiao, when analyzed as a corpus rather than as individual texts, reveal systematic patterns—a 57.7% TRI-UMPH dominance, a 19.2% RESILIENCE minority, a 0.0% ASCETIC absence, and a 0.69 mean positive affect—that are reproducible, comparable, and auditable. Narrative arc classification (NAT, MI4) transforms qualitative interpretation of biographical text into a computable, queryable feature.

Second, text-derived features can be operationalized as a quantitative indicator system. The five-metric framework (UD, MBI, CRC, NAT, WER) provides a computational dashboard applicable to any unstructured biographical corpus: which texts contain unverified claims (high UD), which temporal periods are missing (negative MBI), which texts corroborate each other (high CRC), which narrative types are over- or under-represented (NAT distribution), and which

texts offer the greatest enrichment to public knowledge graphs (high WER). These metrics are cheap to compute, interpretable, and directly actionable for downstream applications.

Third, domain-specific extraction complements public knowledge graphs (RQ2). Wikidata provides authoritative identity anchors for millions of entities but averages only 1.8 facts per scientist in our cohort and contains zero relational statements (mentorship, collaboration, life events). LLM extraction from nianbiao produces $36.8\times$ more facts, filling precisely the relationship types that general-purpose KGs lack. The two are architecturally complementary: Wikidata identity backbone + domain extraction relational layer = rich, queryable structured biographical data.

Fourth, the enrichment effect is quantifiable and supports iterative re-assessment (RQ3). Publishing our 26-scientist extraction to Wikidata would contribute approximately 1,717 net-new linked data statements at a cost of approximately \$0.0007 per fact. After enrichment, the MI1–MI5 indicators can be recomputed to assess whether extraction quality improved—for instance, whether CRC rises from 0.0% as enriched Wikidata links create cross-text connections, or whether UD decreases as public data corroboration resolves uncertain claims. Three categories of content remain irreducibly scarce even after maximum enrichment: personal reflections never formally published, tacit experimental knowledge, and informal mentor networks—findings that guide where further text acquisition, rather than automated enrichment, is needed. This closed-loop framework—assess → enrich → re-assess → follow up—transforms text extraction quality assessment from a one-time step into an iterative, computationally-driven process.

The practical contribution is a shift from passive text archiving to proactive, computationally-driven extraction and quality assessment. The pathway—LLM extraction → indicator dashboard (MI1–MI5) → Wikidata-aligned identity resolution → linked data publication—is technically demonstrated, quantitatively validated at 97.5% precision, and economically feasible at scale. The methodology is replicable across languages and text types, offering a general framework for computational extraction of unstructured biographical text that bridges domain-specific corpora and the global knowledge graph ecosystem.

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References

- [1] De Paiva, V., & Rademaker, A. (2024). Towards a Brazilian history knowledge graph. *arXiv preprint*, 2403.19856.
- [2] Doerr, M. (2003). The CIDOC conceptual reference module. *AI Magazine*, 24(3), 75–92.
- [3] Dooley, J. M., & Luce, K. (2010). *Taking our pulse*. OCLC Research.
- [4] Gottschalk, S., & Demidova, E. (2019). EventKG—the hub of event knowledge on the Web—and biographical timeline generation. *Semantic Web*, 10(6), 1039–1070.
- [5] Griffin, M. (2014). From text to network: Computational approaches to historical biography. *Digital Scholarship in the Humanities*, 29(4), 711–723.
- [6] Hunter, M. (Ed.). (2017). *Archives of the scientific revolution*. Boydell & Brewer.
- [7] Kwaśnik, B. H. (2019). Knowledge organization in the archival domain. *Knowledge Organization*, 46(7), 505–517.
- [8] Larson, R. R., & Janakiraman, K. (2015). Connecting archival collections: The SNAC Project. In *TPDL 2015* (pp. 3–14). Springer.
- [9] Liu, C.-L., & Wang, H. (2017). Matrix and graph operations for relationship inference: An illustration with the kinship inference in the China Biographical Database. In *Proc. 2017 Int. Conf. Digital Humanities* (pp. 1–5).
- [10] O’Neill, E. T., & Casserly, M. F. (2019). Special collections in academic libraries. *College & Research Libraries*, 80(2), 189–204.
- [11] Pan, S., Luo, L., Wang, Y., Chen, C., Wang, J., & Wu, X. (2023). Unifying large language models and knowledge graphs: A roadmap. *IEEE Trans. Knowledge and Data Engineering*, 36(7), 3125–3145.
- [12] Rubin, D. C., Wetzler, S. E., & Nebes, R. D. (1986). Autobiographical memory across the adult lifespan. In D. C. Rubin (Ed.), *Autobiographical memory* (pp. 202–221). Cambridge U. Press.
- [13] Sherratt, T. (2020). The biographical turn in digital humanities. In Bode & Arthur (Eds.), *Advancing Digital Humanities* (pp. 97–114). Palgrave.
- [14] Trace, C. B., & Dillon, A. (2012). The evolution of the finding aid. *Archival Science*, 12(4), 501–519.

- [15] Trace, C. B., & Karadkar, U. P. (2020). Information management in the humanities. *JASIST*, 71(4), 445–459.
- [16] Twitchett, D. C. (1962). Chinese biographical writing. In Beasley & Pulleyblank (Eds.), *Historians of China and Japan* (pp. 95–114). OUP.
- [17] Wilkinson, E. (2018). *Chinese history: A new manual* (5th ed.). Harvard U. Press.
- [18] Zhang, L., & Liu, W. (2018). Scientific archives in Chinese university libraries. *J. Academic Libraries*, 36(3), 45–52.
- [19] Lu, Y., Wu, W., Zhao, X., & Peng, R. (2025). KARMA: Leveraging multi-agent LLMs for automated knowledge graph enrichment. *arXiv preprint*, 2502.06472.
- [20] Carriero, V. A., Azzini, A., Baroni, I., Scrocca, M., & Celino, I. (2024). Human evaluation of procedural knowledge graph extraction from text with large language models. *arXiv preprint*, 2412.03589.
- [21] Blanke, J., & Riechert, T. (2020). Towards an RDF knowledge graph of scholars from early modern history. *arXiv preprint*, 2009.06337.
- [22] Portelli, A. (1991). *The death of Luigi Trastulli and other stories: Form and meaning in oral history*. SUNY Press.
- [23] Thompson, P. (2000). *The voice of the past: Oral history* (3rd ed.). Oxford University Press.
- [24] Trainin, I., Keydar, R., & Pinchevski, A. (2026). The shape of testimony: A scalable framework for oral history archive comparison. *arXiv preprint*, 2605.21623.
- [25] Reagan, A. J., Mitchell, L., Kiley, D., Danforth, C. M., & Dodds, P. S. (2016). The emotional arcs of stories are dominated by six basic shapes. *EPJ Data Science*, 5(1), 31.
- [26] Jockers, M. L. (2015). *Syuzhet: Extract sentiment and plot arcs from text*. R package version 1.0.4.
- [27] Allison-Cassin, S., & Scott, D. (2018). Wikidata: A platform for your library’s linked data. *The Code4Lib Journal*, (40).